

Unbounded consecutive failures of log-concavity from one fixed three-vertex branch

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Abstract

Let T_m have one root, m children, and two pendant leaves at each child. We give an elementary proof that its independence polynomial has a consecutive interval of strict log-concavity failures of length at least $2 \log \log m / \log(3/2) - O(1)$. The tree has diameter four, and every vertex except the root has degree at most three. Its repeated rooted branch is fixed. We also give an explicit finite sufficient criterion, requiring only integer comparisons. This is an independently derived structural specialization, not a claim of priority for unbounded consecutive failures, which were already established in work of Xiyao Yu.

Scope and prior work. Bautista-Ramos proved unbounded numbers of failures, without requiring consecutive indices [1]. Bautista-Ramos, Guillén-Galván and Gómez-Salgado constructed consecutive runs of specified small lengths [2]. Yu's September 2026 archive establishes unbounded runs and the substantially stronger unrestricted extremal lower bound $M(N) = \Omega(N^{1/3})$ [3]. The present note supplies a short proof for a fixed three-vertex module and diameter-four trees. Its $\log \log N$ bound does not improve Yu's unrestricted growth bound. Priority for this particular specialization has not been established. The argument below is self-contained; it does not resolve the tree unimodality conjecture.

Construction. For an integer $m \geq 2$, the edges of T_m are

$$\{vw_i, w_i a_i, w_i b_i : 1 \leq i \leq m\}.$$

Thus $|V(T_m)| = 3m + 1$, its diameter is four, and $\alpha(T_m) = 2m + 1$. Write $I_m(x) = \sum_k i_k(T_m) x^k$. Conditioning on the root gives

$$I_m(x) = (1 + 3x + x^2)^m + x(1 + x)^{2m}. \quad (1)$$

The reflected polynomial is

$$R_m(x) = x^{2m+1} I_m(1/x) = (1 + x)^{2m} + x(1 + 3x + x^2)^m. \quad (2)$$

Put $a_j = [x^j] R_m = i_{2m+1-j}(T_m)$ and $r = 3/2$.

Theorem 1 (Finite criterion). *If the positive integers m, j satisfy*

$$j \geq 48, \quad \frac{72m}{j^2} \leq r^j \leq \frac{m}{8}, \quad m \geq 16(j+1)^3, \quad (3)$$

then $a_j^2 < a_{j-1} a_{j+1}$.

Lemma 2 (Coefficient bounds). *Define*

$$b_0 = 1, \quad b_j = \frac{(2m)^j}{j!} + \frac{(3m)^{j-1}}{(j-1)!} \quad (j \geq 1).$$

For $0 \leq j \leq m$,

$$\left(1 - \frac{j(j-1)}{2m}\right) b_j \leq a_j \leq b_j. \quad (4)$$

Proof. For $0 \leq k \leq m$,

$$\binom{m}{k} 3^k \leq [x^k](1 + 3x + x^2)^m \leq \frac{(3m)^k}{k!}.$$

The lower bound uses the terms choosing $3x$ from k distinct factors. For the upper bound factor $1 + 3x + x^2 = (1 + \lambda_+ x)(1 + \lambda_- x)$, where $\lambda_{\pm} = (3 \pm \sqrt{5})/2 > 0$. The k th elementary symmetric polynomial in nonnegative weights is at most the k th power of their sum divided by $k!$. Also

$$\binom{m}{k} = \frac{m^k}{k!} \prod_{s=0}^{k-1} (1 - s/m) \geq \frac{m^k}{k!} \left(1 - \frac{k(k-1)}{2m}\right).$$

Apply these bounds with $k = j - 1$, and the analogous bounds to $\binom{2m}{j}$. Their sum gives (4); $j = 0$ is immediate. $\square$

Proof of Theorem 1. For $j \geq 1$ put $t = jr^j/(3m)$, so $b_j = (2m)^j(1+t)/j!$. The adjacent values of this parameter are $t(j-1)/(jr)$ and $t(j+1)r/j$. Direct algebra gives the exact identity

$$\frac{b_{j-1}b_{j+1}}{b_j^2} - 1 = \frac{t}{(j+1)(1+t)^2} \left(\frac{j-7}{6} - \frac{1}{t} - \left(1 + \frac{1}{j}\right)t \right). \quad (5)$$

The middle conditions in (3) imply $24/j \leq t \leq j/24$. Hence, for $j \geq 48$,

$$\frac{j-7}{6} - \frac{1}{t} - \left(1 + \frac{1}{j}\right)t \geq \frac{j}{12} - \frac{29}{24} \geq \frac{j}{24}.$$

On that same interval,

$$\frac{t}{(1+t)^2} \geq \frac{24/j}{(1+24/j)^2} \geq \frac{32}{3j}.$$

Therefore (5) is at least $4/(9(j+1)) \geq 1/(4j)$. Let $\delta = (j+1)^2/(2m)$. Lemma 2, applied at the three indices, now yields

$$\frac{a_{j-1}a_{j+1}}{a_j^2} \geq (1-\delta)^2 \left(1 + \frac{1}{4j}\right) \geq \left(1 - \frac{1}{16(j+1)}\right) \left(1 + \frac{1}{4j}\right) > 1.$$

Here $m \geq 16(j+1)^3$ ensures both the applicability of the lemma and $2\delta \leq 1/(16(j+1))$. This proves the strict inequality. $\square$

Corollary 3 (Unbounded runs with a fixed branch). *For all sufficiently large m , the polynomial I_m has a consecutive interval of strict log-concavity failures of length*

$$\frac{2}{\log(3/2)} \log \log m + O(1).$$

In particular the longest such interval is at least this quantity. The coefficient offsets from degree $2m+1$ lie between

$$\frac{\log m - 2 \log \log m + O(1)}{\log(3/2)} \quad \text{and} \quad \frac{\log m + O(1)}{\log(3/2)}.$$

Proof. Let $\lambda = \log(3/2)$, and let $u_m > 0$ be the unique real solution of $u_m^2 e^{\lambda u_m} = 72m$. Put $v_m = \log(m/8)/\lambda$. Taking logarithms and substituting back gives

$$u_m = \frac{\log m - 2 \log \log m + \log 72 + 2 \log \lambda + o(1)}{\lambda}, \quad v_m = \frac{\log m - \log 8}{\lambda}.$$

Every integer $j \in [\lceil u_m \rceil, \lfloor v_m \rfloor]$ satisfies the middle inequalities of (3), since $j^2 r^j$ is increasing. For sufficiently large m the interval is nonempty, its lower endpoint is at least 48, and $m \geq 16(\lfloor v_m \rfloor + 1)^3$. Theorem 1 applies throughout. The number of integers in the interval equals $(2/\lambda) \log \log m + O(1)$. Reflection reverses the interval but preserves the strict log-concavity inequalities. $\square$

Why the effect persists. The two terms in (2) have coefficients approximately $(2m)^j/j!$ and $(3m)^{j-1}/(j-1)!$. Their ratio is $j(3/2)^j/(3m)$. The crossover moves to offsets of order $\log m$, where the log-concavity of the factorial denominator has size $1/j$. The positive curvature from the mixture exceeds that size over an interval of width proportional to $\log j$. Formula (5) and the finite bounds make this explanation rigorous.

Exact checks. The accompanying standard-library Python program calculates coefficient prefixes with arbitrary-precision integers. The displayed intervals were checked directly using $a_{j-1}a_{j+1} - a_j^2 > 0$; no floating-point sign checks are used. The intervals are maximal within the computed prefix, not a claim to enumerate all failures in the full polynomial.

m	Reflected failure offsets	Consecutive length
2^{16}	22–24	3
2^{32}	44–52	9
2^{64}	95–107	13
2^{128}	200–217	18
2^{256}	416–435	20

Writing $p_k = [x^k](1 + 3x + x^2)^m$ gives the exact recurrence

$$p_0 = 1, \quad p_1 = 3m, \quad (k+1)p_{k+1} = 3(m-k)p_k + (2m-k+1)p_{k-1}.$$

Then $a_0 = 1$ and $a_j = \binom{2m}{j} + p_{j-1}$. The verifier also compares this recurrence against direct polynomial multiplication on small trees, and checks the finite criterion independently of the observed intervals.

Status. The mathematical claim is exactly Theorem 1 and its corollary, with the proof printed above. The numerical checks are supplementary. This draft has not undergone independent expert review. The unrestricted existence theorem belongs to the prior literature; novelty of the fixed-module, diameter-four specialization remains to be checked beyond the sources consulted. The failures occur in the far tail and do not imply non-unimodality.

References

- [1] C. Bautista-Ramos, *Multiple breaks of log-concavity in the independence polynomials of trees*, arXiv:2511.00334 (2025). <https://arxiv.org/abs/2511.00334>.
- [2] C. Bautista-Ramos, C. Guillén-Galván and P. Gómez-Salgado, *Linear recurrences for non-log-concave independence polynomials of trees*, *Graphs and Combinatorics* **42**, 59 (2026). <https://doi.org/10.1007/s00373-026-03054-4>.
- [3] X. Yu, *Unbounded runs of non-log-concavity in independence polynomials of trees*, manuscript in reproducibility archive, version 1.1.0, 3 September 2026. <https://doi.org/10.5281/zenodo.22263331>.